EEL 6878 Homework 4

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Question 1.

The plot below shows the Graph Fourier Transform of the Heat filtered random signals.

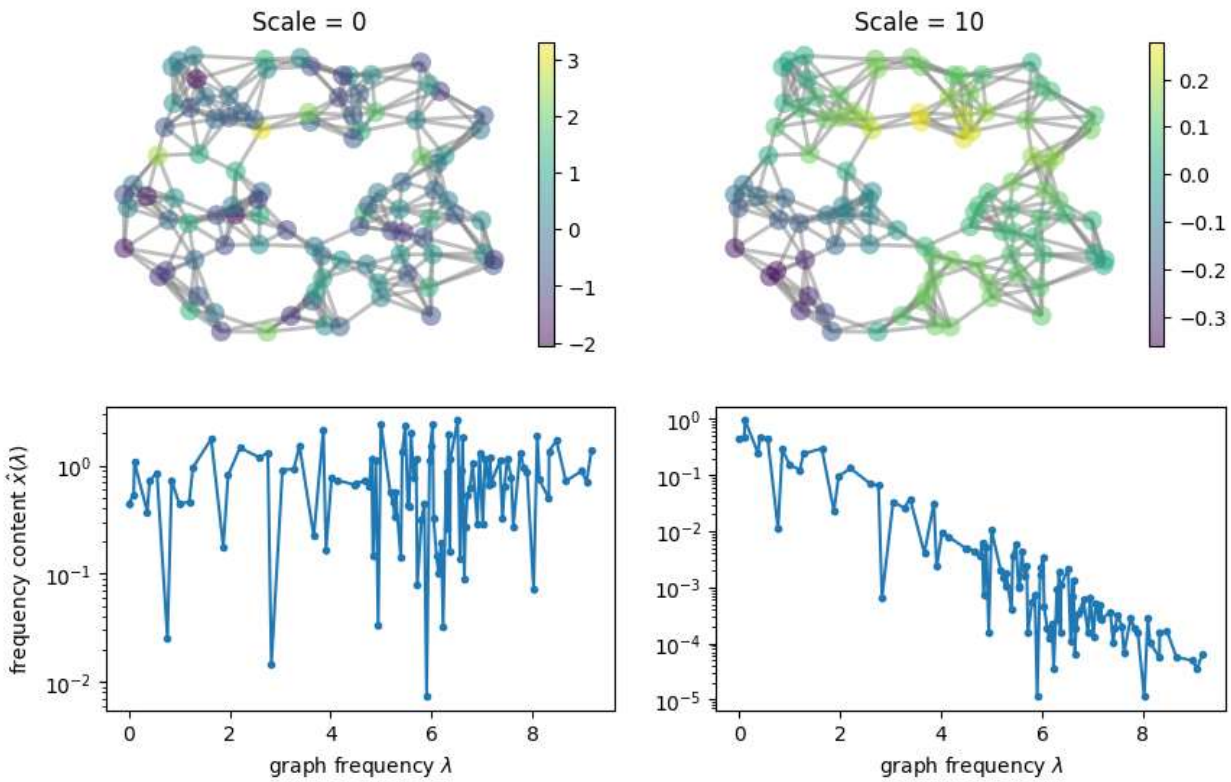

Figure 1: GFT of Heat Filtered Random Graph Signals

The graph signal generated by applying a Heat filter bank with $\text{scale} = 10$ is a low-pass signal. Low frequencies see little to no attenuation, while high frequencies see reduction in power by orders of magnitude.

When applying the Heat kernel with $\text{scale} = 0$ to the random signal, the resulting signal had a smoothness of 580.9577. For the case when $\text{scale} = 10$, the resulting signal had a smoothness of 0.7838.

The higher scale, low-pass filter produced a significantly smoother signal, which can be observed in the top row of Figure 1. Transitions between node values at $\text{scale} = 10$ appear much more gradual than for $\text{scale} = 0$. This is expected behavior. Higher frequencies present in the signal are responsible for more drastic changes in the values of connected nodes. This

allows for a rougher signal with more variation. Attenuating the signal at these frequencies removes most of the potential for large changes between nodes. A low-pass filter attenuates or eliminates high-frequency content inherently. The Heat filter with $\text{scale} = 10$ does more to eliminate higher frequencies in the signal, as seen in the second row of Figure 1. Higher frequency content remains present in the filtered signal for $\text{scale} = 0$ while being attenuated by orders of magnitude when $\text{scale} = 10$, hence a rougher signal.

Question 2.

Part 1) Loaded Graphs and Their Eigenvalues

The Erdős-Rényi graph has frequencies that increase steadily. The Ring graph has frequencies which follow that of a circulant graph. The graph created by the Cora dataset has many very low frequencies. From the plots, the Cora graph has several connected components, meaning several of the Laplacian eigenvalues will be 0. The several other low frequencies indicate the graph or its components have low connectivity. This makes sense from how many nodes and components the graph and dataset contain.

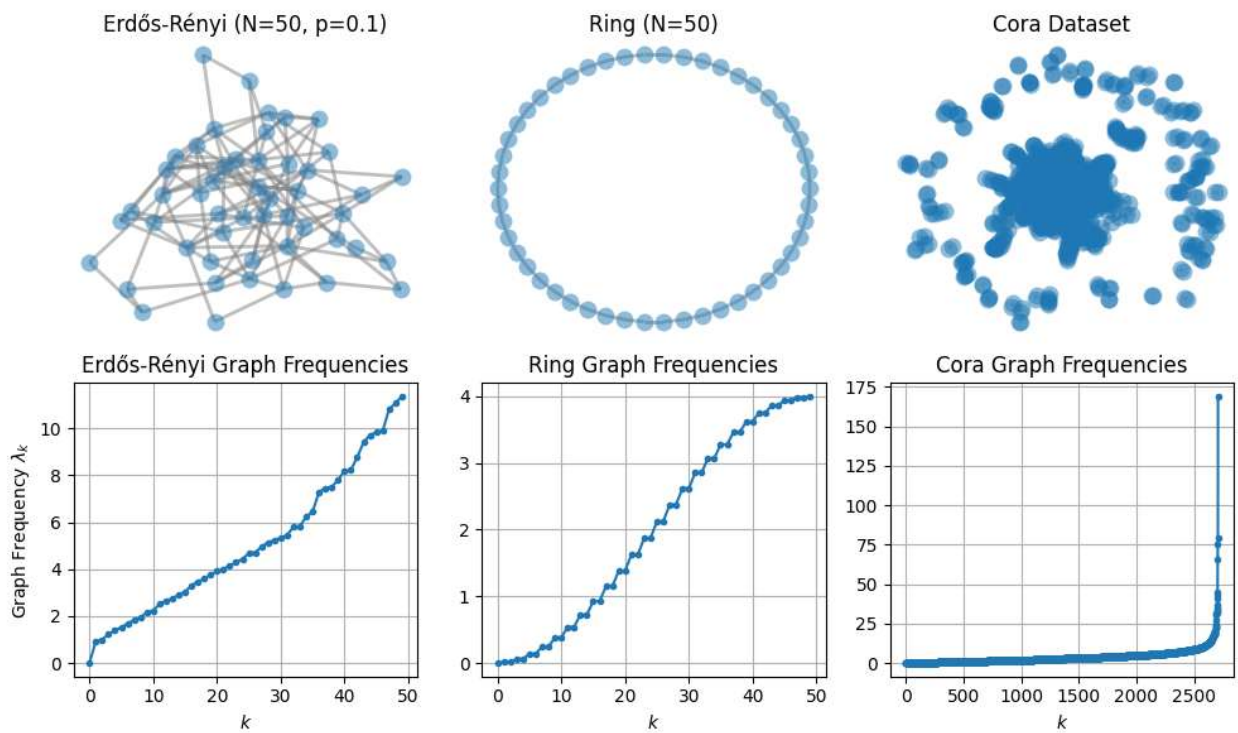

Figure 2: Generated or Loaded Graphs and Spectral Content

Part 2) Graph Eigenvectors

The Cora graph contains several connected components, so $\lambda_2 = \lambda_3 = \lambda_6 = 0$. Each of the eigenvectors correspond to a different connected component set of components. From Figure 2 and Figure 3, it seems the graph of the Cora dataset has a rather large nullity, and many such eigenvalue/eigenvector pairs can be expected.

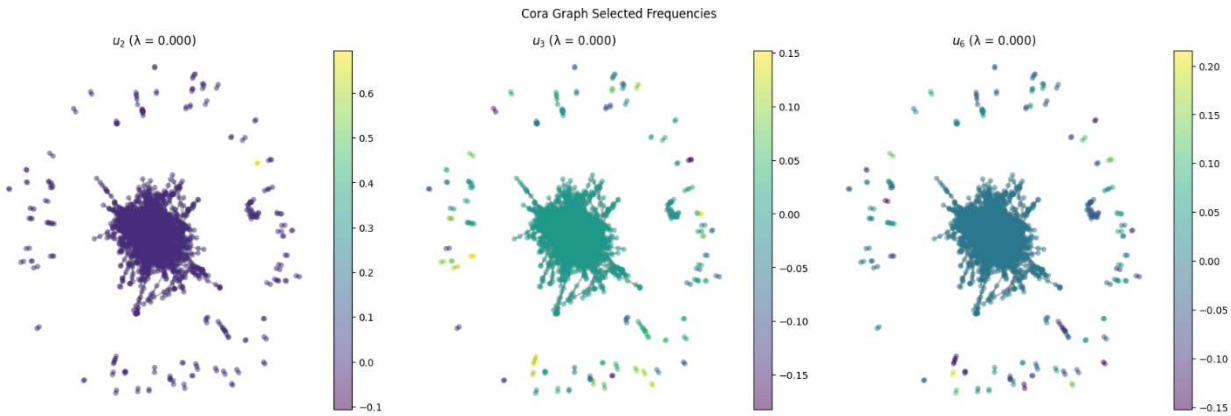

Figure 3: Cora Graph Eigenvectors of Selected Frequencies

Embedding and Labelling

After performing the t-SNE and applying K-means clustering for labeling, the resulting labels achieved an ARI of 0.0928 and a Silhouette score of 0.364. The Silhouette score indicates that the K-means clustering is able to find a decent geometry for the K-means clustering within the embedding space. From the ARI scores, there is very little significance

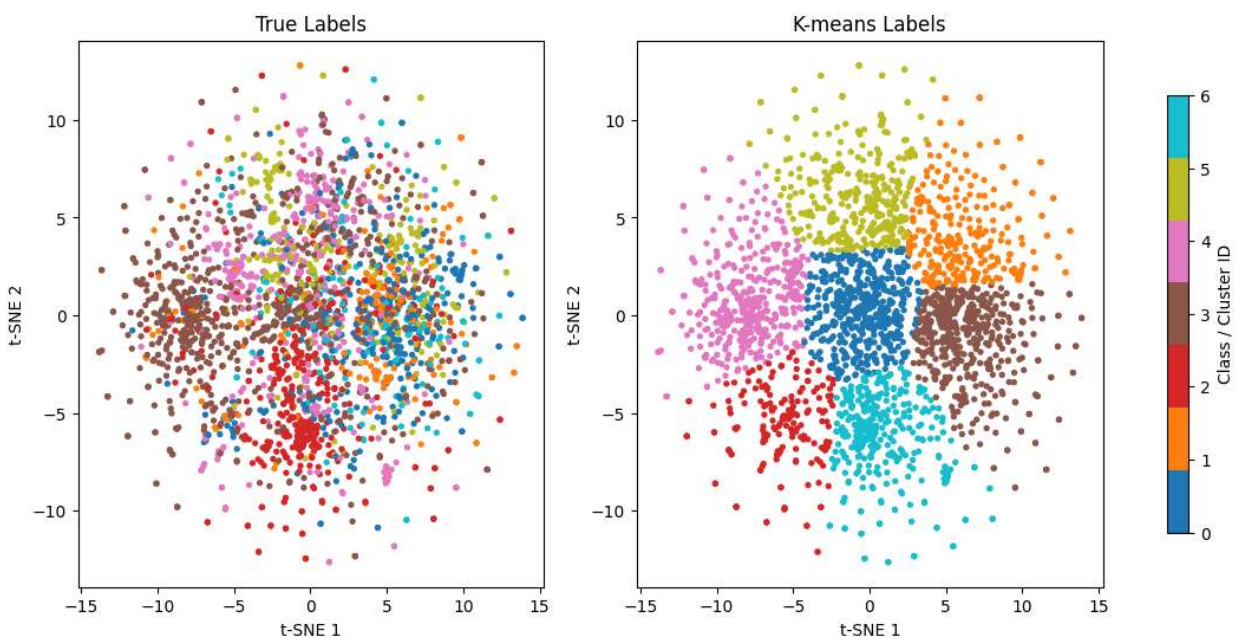

Figure 4: t-SNE + K-means Label Solution

in the K-means clustering concerning the true labeling of nodes. Instead, the near zero ARI indicates the K-means labeling solution is little better than random chance. These scores reveal that the 2-component t-SNE embedding is, at best, a somewhat decent selection of an embedding space selection with which to apply K-means. *Figure 4* shows the true labels of nodes and the node labels according to the K-means solution. These nodes just aren't well neatly grouped within the embedding space. Instead, they are well mixed. Either a higher dimensional embedding may be needed, or a different clustering technique could be used.

Ring Signal Denoising

The noisy signal in the Ring graph and its spectral content are shown below. Low frequency content is prominent due to the construction of the original signal. The variation in random noise is represented in the signal by high frequency components. The original signal was constructed with periodic functions, but many of the node values show large deviations from periodic behavior. Instead, there are large swings such as from 1.5 to 1.0 or -1.5 to -2.5 and back.

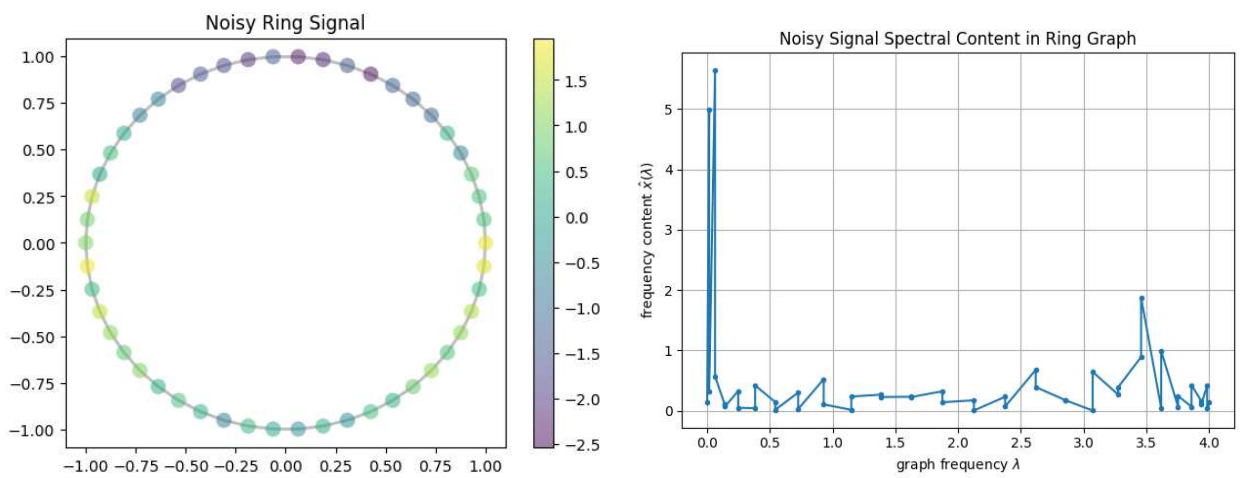

Figure 5: Noisy Signal Results

Denoised Signal

Compared to the noisy signal, the filtered and denoised signal has a much more prominent periodic component present. Figure 6 shows the results of applying a low-pass filter to the graph signal. The frequency content for larger graph frequencies has been greatly reduced. In attenuating the high frequency signal components, the low-pass filter

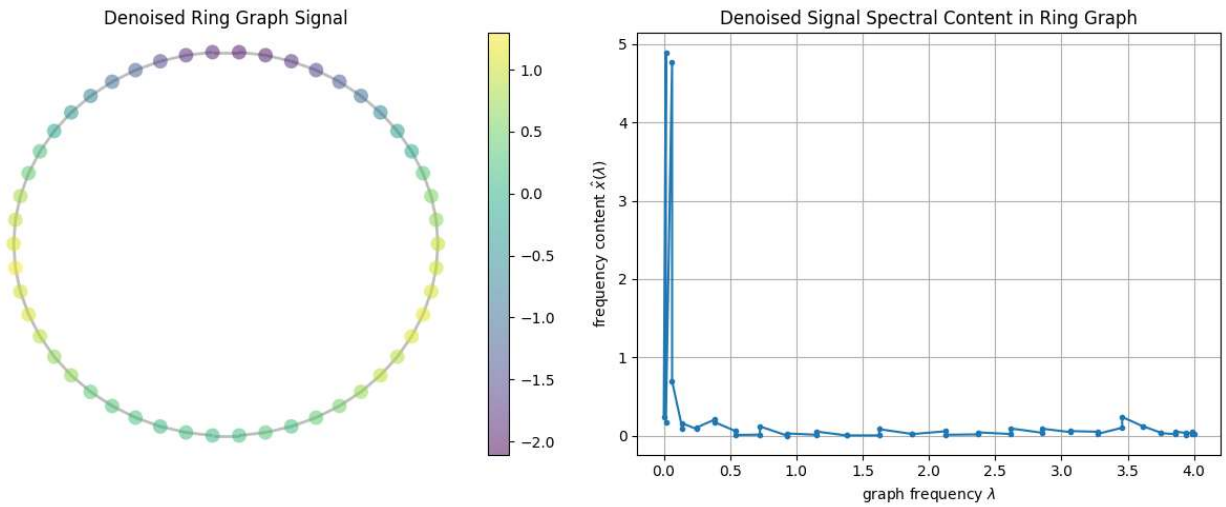

Figure 6: Denoised Results

significantly reduces, but does not eliminate, the contribution of random noise to the graph signal.

Question 3.

Training Loss

The plot below shows the training loss across 200 training epochs. The training loss shows a steep decline towards zero, followed by a plateau at lower magnitudes. The model starts off with roughly uniform statistical likelihood for node labels. It seems that most improvements are made in early epochs. At later epochs, the model is making incremental improvements.

Model Evaluations

The plot below shows the node labels according to the Cora dataset, the labels as predicted by the ChebNet, and the disagreement between the two values. The smaller blue nodes are nodes for which the labels agree. Larger red nodes are nodes for which the labels disagree. The ChebNet is able to correctly predict the labels for a vast majority of the nodes. Overall, the model achieved 74.2% accuracy. The ChebNet produces a significantly better estimate than a t-SNE embedding and clustering based approach.

From the label disagreement, the outer clusters seem particularly mislabeled by the ChebNet. This is caused by the isolation of these clusters, so these nodes are not likely to share edges with training nodes. Thus, the model has little ability to spread this information across the graph. The central cluster contains many more connections, so nodes will see much more interaction between labels. The interconnectivity may be a reason the label prediction performs as well as it does. Information and features from the training nodes will be able to propagate to their neighbors and thus more nodes.

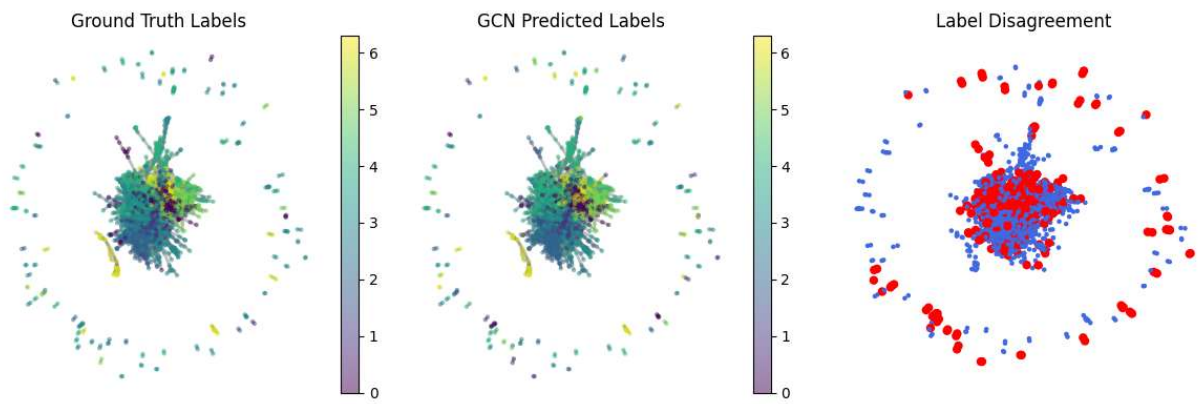

Figure 7: ChebNet and Truth Labels

Code

The code used for this project is stored at
https://github.com/QuiteFiber93/eel6878/tree/main/hw4